Remainder as Observable

Exact Quotient-Remainder Decomposition Across Six Domains of Quantum Physics

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I. ABSTRACT

This paper presents three independent contributions.

First, an observational claim: in five domains of quantum physics — Berry phase, Brillouin zones, Bohr-Sommerfeld quantization, Chern-Simons theory, and QCD theta vacuum — physical quantities decompose formally into an integer quotient and a remainder under division by a domain-specific modulus. The integer is topologically protected and cannot change continuously. The remainder is the physical observable. A sixth domain, renormalization group flow, has an analogous but not formally integer structure. Each decomposition is individually established in the literature. The contribution here is the unified observation that all six share a common pattern: integer = quantum number, remainder = observable, modulus = symmetry.

Second, a computational claim: the Q335 exact arithmetic framework from [HOWL-MATH-4-2026], where transcendental constants are single integers over a shared denominator 2^{335}, makes the quotient-remainder decomposition exact at every computational step. When a physical quantity represented in the Q335 basis is divided by a basis constant, both the integer quotient and the remainder are exact ratios of integers. No rounding occurs. This is demonstrated for the electromagnetic chain from [HOWL-PHYS-9-2026], where the QED series inversion and vacuum polarization running produce exact remainders at each threshold.

Third, a data claim: a systematic search for the modulus that extends this structure to the Standard Model coupling constants returns null. Seventeen measured SM dimensionless ratios tested against 34 transcendental basis constants as moduli, with PSLQ decomposition of residuals at maxcoeff = 10,000, produce no clean integer quotients. The modulus

connecting the integer arithmetic framework to SM parameter values is not any single basis constant or simple rational fraction thereof. This bounds the search space for future work but does not close it.

Each claim is independent. Claim 1 stands without Claims 2 or 3. Claim 2 stands without Claim 3. Claim 3 is the honest report of a negative search. No SM parameter is derived. No modulus is identified. The structure is proven, the tool is demonstrated, and the search is reported.

II. FIVE FORMAL DOMAINS

In each of the following five domains, a physical quantity Q decomposes under division by a modulus M into an integer quotient $q = \lfloor Q/M \rfloor$ and a remainder $R = Q - q \cdot M$. The quotient is topologically protected. The remainder is the observable.

2.1 Berry Phase

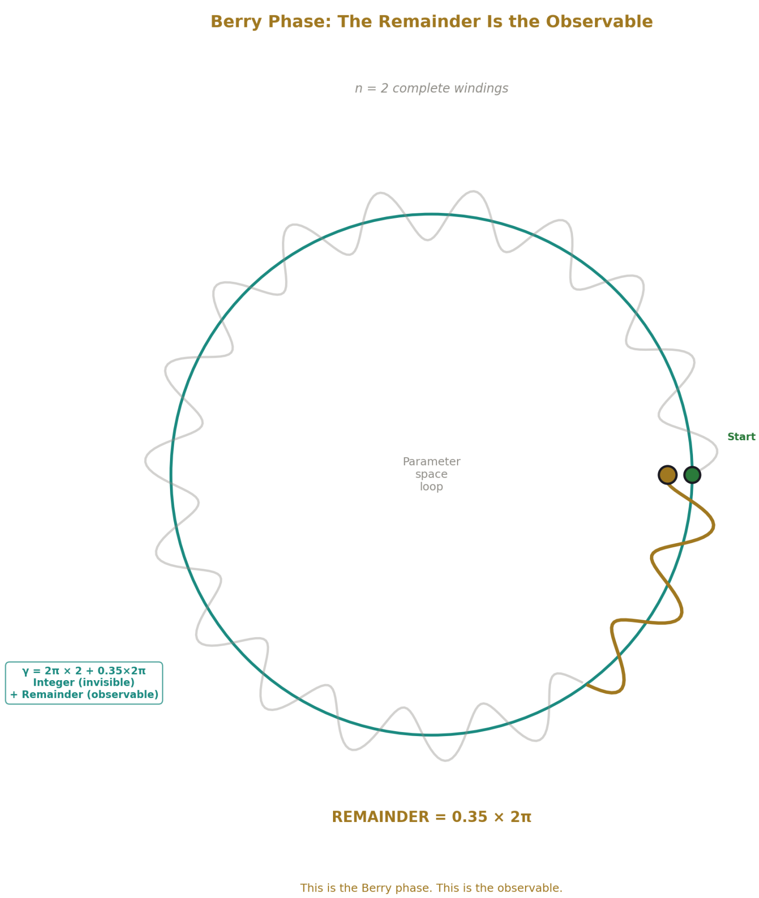

Figure 1: Fig. 5: A quantum system transported around a parameter space loop accumulates integer windings (invisible) plus a fractional remainder (the Berry phase) — the remainder IS the observable.

A quantum system transported adiabatically around a closed loop in parameter space acquires a geometric phase γ in addition to the dynamical phase. The geometric phase decomposes:

$\gamma = 2\pi \cdot n + (\gamma \bmod 2\pi)$

The integer n is the winding number — the number of complete phase cycles accumulated. It is gauge-dependent and physically invisible. The remainder $\gamma \bmod 2\pi$ is gauge-invariant and physically observable. It determines the Hall conductance $\sigma\_xy = (e^2/h)\cdot C$ through the Chern number $C = (1/2\pi)\oint\Omega\cdot dS$, where Ω is the Berry curvature.

Component	Identification
Quantity	Geometric phase $\gamma = i\oint \langle \psi\|\nabla R\|\psi \rangle \cdot dR$
Modulus	2π (full phase cycle)
Integer	Winding number n
Remainder	γ mod 2π (geometric phase)
Observable	Hall conductance, Aharonov-Bohm fringe shift
Topological protection	n changes only by discrete integer jumps (phase transition required)
Source of modulus	Periodicity of the complex exponential e^(iγ)
Established by	Berry 1984, Simon 1983, Thouless Kohmoto Nightingale den Nijs 1982

2.2 Brillouin Zone

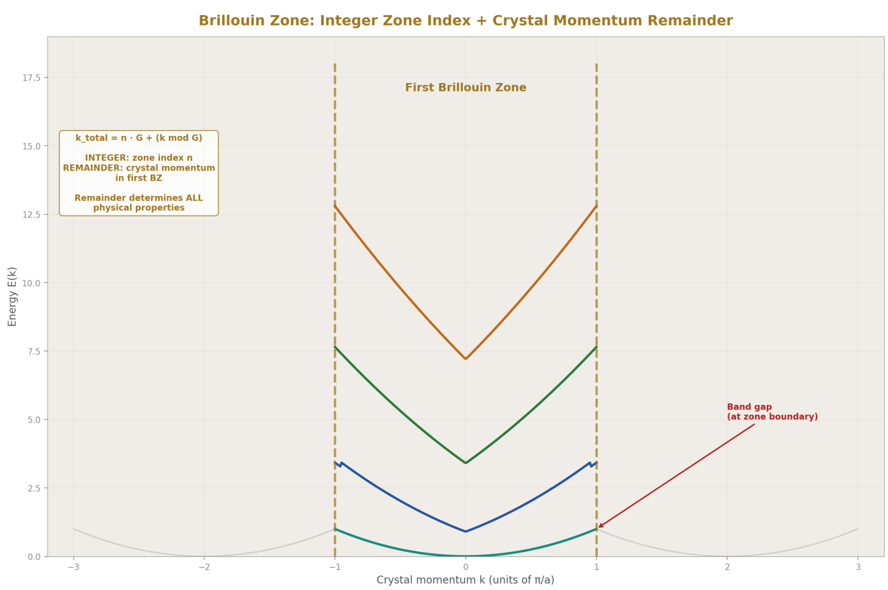

Figure 2: Fig. 2: Band structure E(k) with zone boundaries — crystal momentum folds into the first BZ. Zone index is the integer, k mod G is the remainder that determines all physical properties.

An electron in a periodic crystal has crystal momentum k defined only modulo the reciprocal lattice vector $G = 2\pi/a$, where a is the lattice constant. The momentum decomposes:

k_total = n·G + (k mod G)

The integer n identifies which Brillouin zone. It is a labeling convention. The remainder k mod G — the crystal momentum within the first Brillouin zone — determines all physical properties: band structure E(k), group velocity ∂E/∂k, effective mass, conductivity. Band gaps open at the zone boundary where k = G/2, which is the point of maximum remainder (remainder = modulus/2).

Component	Identification
Quantity	Crystal momentum k
Modulus	$G = 2\pi/a$ (reciprocal lattice vector)
Integer	Zone index n
Remainder	k mod G (crystal momentum in first BZ)
Observable	Band structure, conductivity, band gaps
Topological protection	Zone index changes only by discrete umklapp scattering
Source of modulus	Discrete translation symmetry of the lattice
Established by	Bloch 1929, Brillouin 1930

2.3 Bohr-Sommerfeld Quantization

The classical action around a closed orbit is quantized in units of $2\pi\hbar$, with a fractional correction from the Maslov index:

$$\oint p \cdot dq = 2\pi\hbar \cdot (n + \mu/4)$$

The integer n is the quantum number counting de Broglie wavelengths in the orbit. The remainder $\mu/4$ is the Maslov correction, determined by the number of classical turning points (caustics) encountered on one circuit. For the harmonic oscillator, there are two turning points ($\mu = 2$), giving the zero-point energy $\hbar\omega/2$. For a particle bouncing between a hard wall and a soft turning point, $\mu = 3$, giving correction 3/4.

Component	Identification
Quantity	Classical action $\oint p \cdot dq$
Modulus	$2\pi\hbar$
Integer	Quantum number n
Remainder	$\mu/4$ (Maslov index)
Observable	Energy level correction, zero-point energy $E_0 = \hbar\omega/2$
Topological protection	n changes only at eigenvalue crossings; μ changes only when orbit topology changes
Source of modulus	de Broglie relation and single-valuedness of the wavefunction
Established by	Bohr 1913, Sommerfeld 1916, Maslov 1965, Keller 1958

The Maslov correction is itself a rational number. In exact Fraction arithmetic, the corrected energy level $E_n = \hbar\omega(n + 1/2)$ involves no irrational or transcendental quantity — it is an integer plus 1/2 times a measured constant. The remainder is exactly representable.

2.4 Chern-Simons Theory

The Chern-Simons functional on a 3-manifold evaluates to a real number whose integer part is the Chern number of the bounding 4-manifold and whose fractional part determines the topological phase:

CS(A) = (Chern number) + (CS mod $\mathbb{Z}$)

The integer part — the Chern number — is a topological invariant. It cannot change under smooth deformations of the gauge field. The fractional part CS mod $\mathbb{Z}$ determines the partition function phase $Z = e^{2\pi i \cdot k \cdot CS}$, and thereby determines the fractional statistics of excitations. For the fractional quantum Hall effect at filling $\nu = p/q$, the CS level $k = q$ and the fractional part encodes the anyonic exchange phase.

Component	Identification
Quantity	CS invariant $CS(A) = (k/4\pi)\int Tr(A\wedge dA + 2A\wedge A\wedge A/3)$
Modulus	1 (from large gauge transformations, which shift CS by integers)
Integer	Chern number
Remainder	CS mod $\mathbb{Z}$ (fractional CS invariant)
Observable	Fractional statistics, anyonic exchange phase, FQHE filling
Topological protection	Chern number is a topological invariant of the 4-manifold
Source of modulus	Gauge invariance under large gauge transformations
Established by	Chern and Simons 1974, Witten 1989, Wen 1990

2.5 QCD Theta Vacuum

The QCD vacuum is characterized by a continuous parameter $\theta \in [0, 2\pi)$ labeling a superposition of topological sectors with different instanton numbers:

$|\theta\rangle = \Sigma_n e^{in\theta} |n\rangle$ The instanton number $n \in \mathbb{Z}$ is the integer — it counts the topological winding of the gauge field. The vacuum angle θ is the remainder. It determines the strength of CP violation in strong interactions through the vacuum energy $E(\theta) = E_0 - \chi_{top} \cdot \cos(\theta)$.

Component	Identification
Quantity	Vacuum angle θ
Modulus	2 π (periodicity of the vacuum energy)
Integer	Instanton number $\nu \in \mathbb{Z}$
Remainder	θ ∈ [0, 2 π)
Observable	CP violation strength in strong interactions
Topological protection	Instanton number is an integer topological invariant of the gauge field
Source of modulus	$\mathbb{Z}$ -periodicity of the SU(3) gauge group's third homotopy: π_3(SU(3)) = $\mathbb{Z}$
Established by	't Hooft 1976, Jackiw and Rebbi 1976, Callan Dashen Gross 1976

This is the one domain where the remainder has been derived within this series. [HOWL-PHYS-7-2026] showed that the energy functional E(θ) = E_0 − χ_top·cos(θ), defined on the $\mathbb{Z}$ -periodic domain, has its unique minimum at θ = 0. The remainder is exactly zero, forced by energy minimization on the integer-periodic domain. The derivation used only the properties of cosine (even, unique maximum at 0, periodic) and the $\mathbb{Z}$ -topology of the instanton vacuum.

2.6 Summary

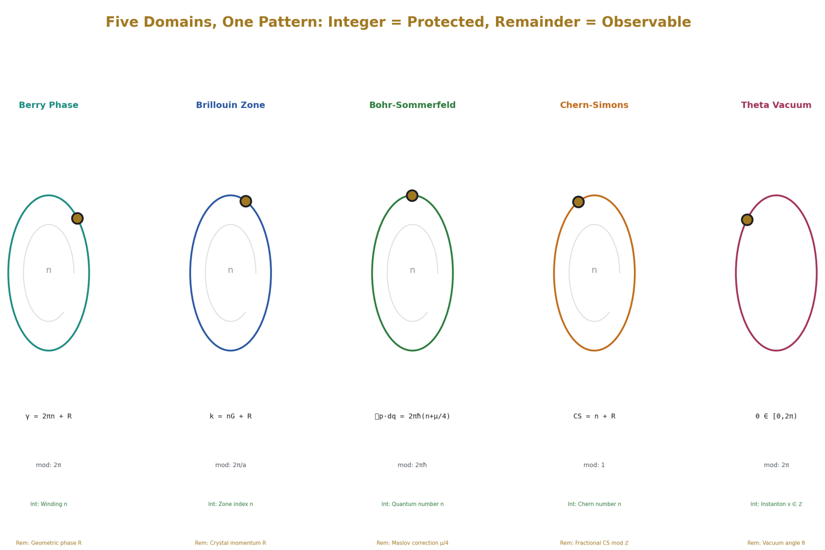

Figure 3: Fig. 3: Berry phase, Brillouin zone, Bohr-Sommerfeld, Chern-Simons, and theta vacuum — each decomposed into integer (protected) plus remainder (observable) under a domain-specific modulus.

Domain	Modulus	Integer	Remainder	Source of modulus
Berry phase	2π	Winding number	Geometric phase	Phase periodicity
Brillouin zone	$2\pi/a$	Zone index	Crystal momentum	Lattice translation symmetry
Bohr-Sommerfeld	$2\pi\hbar$	Quantum number n	Maslov correction $\mu/4$	Action quantization
Chern-Simons	1	Chern number	CS mod $\mathbb{Z}$	Large gauge invariance
Theta vacuum	2π	Instanton number	θ	$\pi_3(SU(3)) = \mathbb{Z}$

In every case: the modulus is determined by a symmetry or topology of the physical system. The integer cannot change continuously — a discrete transition (phase transition, umklapp scattering, eigenvalue crossing, gauge transformation, instanton tunneling) is required. The remainder is the continuously variable degree of freedom within a topological sector. The remainder IS the physics.

III. ONE ANALOGOUS DOMAIN

3.1 Renormalization Group Flow

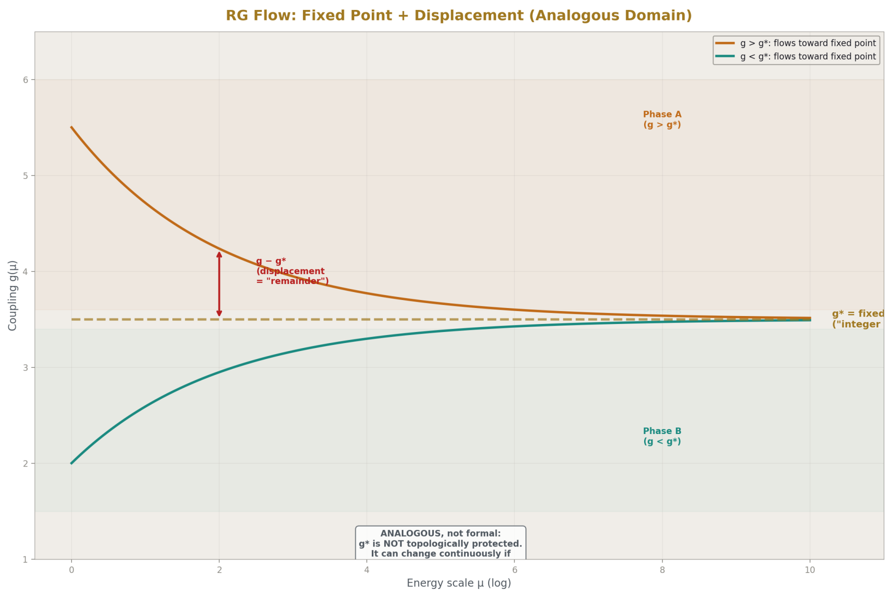

Figure 4: Fig. 6: Coupling g(μ) runs toward fixed point g* — decomposed into g* (reference, "integer analog") and displacement g–g* (remainder). Analogous but not formally integer-protected.

The RG decomposition is structurally analogous to the five formal domains but does not have a literal integer quotient.

The coupling constant g(μ) at energy scale μ evolves according to the beta function β(g) = μ ∂g/∂ μ . At a fixed point g* where β(g*) = 0, the coupling decomposes as:

g(μ) = g* + (g(μ) − g*)

The fixed point g* plays the role of the integer — it is the stable reference value. The displacement g − g* plays the role of the remainder — it determines whether the coupling is growing or shrinking, and thereby determines the phase of matter.

Component	Identification
Quantity	Coupling g(μ) at scale μ
Integer analog	Fixed point g* where β(g*) = 0
Remainder analog	Displacement g − g* from fixed point
Observable	Sign and magnitude determine coupling flow direction, phase of matter

Component	Identification
Key difference	g* is NOT generally an integer. It is a continuous value determined by dynamics

This section is explicitly flagged as analogous. The fixed point plays the role of the integer but lacks topological protection in the sense of the other five domains. The fixed point value can change continuously if the theory's content changes (e.g., varying the number of flavors shifts g* continuously). In the five formal domains, the integer is locked by topology and can only change by discrete jumps.

What makes the RG analog non-trivial: the fixed point is dynamically distinguished (it is a zero of β, not an arbitrary reference value), and the physics on either side of it is qualitatively different (asymptotic freedom vs confinement, or ordered vs disordered phase). The decomposition into "reference + displacement" is physically meaningful, not merely algebraic.

3.2 The VP Running as a Concrete Example

The vacuum polarization running from [HOWL-PHYS-5-2026] provides a concrete instance of the RG analog. The coupling α^{-1} runs between energy scales. At each fermion mass threshold, a discrete event occurs: a new fermion becomes active in the vacuum polarization. Between thresholds, the running is continuous (logarithmic).

The discrete threshold crossings count fermions (an integer). The continuous running between thresholds is the remainder. The Q335 framework computes both exactly.

IV. EXACT REMAINDER TRACKING IN Q335

4.1 The Arithmetic

In the Q335 basis ([HOWL-MATH-4-2026]), every transcendental constant C is represented as an integer numerator p_C divided by the shared denominator 2^{335}:

C = p_C / 2^{335}

When a physical quantity X = p_X / 2^{335} is divided by a basis constant C, the quotient and remainder are:

q = ⌊p_X / p_C⌋ (exact integer division on the numerators)

r = p_X − q · p_C (exact integer subtraction on the numerators)

The physical quotient is q. The physical remainder is r / 2^{335}. Both are exact — the quotient is an integer and the remainder is an exact rational. No rounding occurs at any stage.

This is the operational meaning of "exact remainder tracking": the quotient-remainder decomposition that defines the physics in the five formal domains is performed as integer division on 100-digit numerators. The 2^{335} denominator is shared and cancels. The computation is integer arithmetic.

4.2 Two Kinds of Remainder in the Electromagnetic Chain

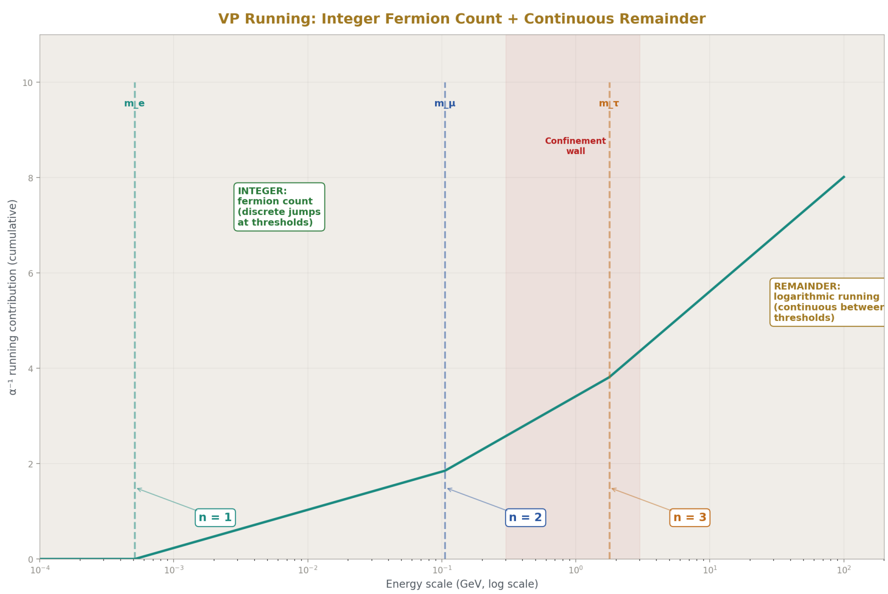

Figure 5: Fig. 1: Fermion count increments at each mass threshold (integer, discrete), VP running accumulates between thresholds (remainder, continuous) — the quotient-remainder decomposition in the electromagnetic chain.

The electromagnetic chain from [HOWL-PHYS-9-2026] — a_e → α → α(M_Z) — produces two structurally distinct kinds of remainder. The paper distinguishes them explicitly.

Type A — Computational residual. The Newton iteration for α from a_e produces a residual $f(x) < 10^{-46}$ at convergence. This measures how well the 4-loop truncated QED series approximates the all-orders series. It is physically meaningful (it bounds missing higher-order contributions) but it is not a quotient-remainder decomposition of a physical quantity against a modulus. It is a convergence measure of an iterative algorithm.

Type B — Physical threshold decomposition. At each lepton mass threshold, the VP running receives a discrete contribution from the newly active fermion. Between thresholds, the running accumulates continuously. The decomposition:

Energy range	Fermion count (integer)	Accumulated running (remainder)
Below m e	0 active leptons	No leptonic VP contribution
m e to m μ	1 active lepton	$\ln(M\ Z^2/m\ e^2)$ accumulated, integer coefficients
m μ to m τ	2 active leptons	$\ln(m\ \mu^2/m\ e^2)$ contribution added
Above m τ	3 active leptons	$\ln(m\ \tau^2/m\ \mu^2)$ contribution added
Hadronic boundary	Perturbative → non-perturbative	Confinement wall from PHYS-6

This IS a formal quotient-remainder decomposition. The fermion count is a non-negative integer. The logarithmic running between thresholds is the continuous remainder. Both are computed exactly in Fraction arithmetic. The integer part (which fermions are active) is protected — it cannot change between thresholds. The remainder (how much running has accumulated) is the observable that determines α at any scale.

V. THE SEARCH

5.1 Method

Two complementary searches were performed. They test different relationships and a null on one does not constrain the other.

Modular search. For each of 17 measured SM dimensionless ratios and each of 34 transcendental basis constants M, compute the exact quotient $q = \lfloor \text{target}/M \rfloor$ and the fractional remainder $R/M = (\text{target} - q \cdot M)/M$. Identify cases where R/M is within 0.1% of a simple fraction p/d with $d \leq 30$.

Linear search (PSLQ). For each target, test whether it is a rational linear combination of basis constants with integer coefficients up to 10,000. This was reported in the PSLQ scan notebook: 17 targets × 3 pool sizes × maxcoeff up to 10,000.

The distinction is critical: PSLQ tests whether a quantity is a rational LINEAR combination of basis constants ($c_1 \cdot x_1 + c_2 \cdot x_2 + \ldots = \text{target}$). The modular search tests whether a quantity is an integer MULTIPLE of a basis constant plus a small correction ($q \cdot M + R = \text{target}$). These are fundamentally different relationships. A number can fail the linear test while passing the modular test, or vice versa.

5.2 The Targets

#	Target	Value	Category
1	α EM^{-1} (from a e, PHYS-9)	137.035998583	Coupling
2	α EM^{-1} (CODATA)	137.035999177	Coupling
3	α s(M Z)	0.1180	Coupling
4	$\sin^2\theta$ W	0.23122	Mixing
5	m μ /m e	206.768	Lepton ratio
6	m τ/m e	3477.15	Lepton ratio
7	m τ/m μ	16.817	Lepton ratio
8	m s/m d	20.0	Quark ratio
9	m c/m s	13.597	Quark ratio
10	m b/m c	3.291	Quark ratio
11	m t/m b	41.313	Quark ratio
12	$\sin\theta_{12}$	0.2253	CKM
13	$\sin\theta_{23}$	0.0412	CKM
14	$\sin\theta_{13}$	0.00350	CKM
15	M W/M Z	0.8815	Boson ratio
16	M H/M Z	1.3719	Boson ratio
17	M H/M W	1.5567	Boson ratio

α_EM^{-1} is tested at both the derived value (from the a_e inversion in PHYS-9, exact Fraction) and the CODATA recommended value.

5.3 Modular Search Results

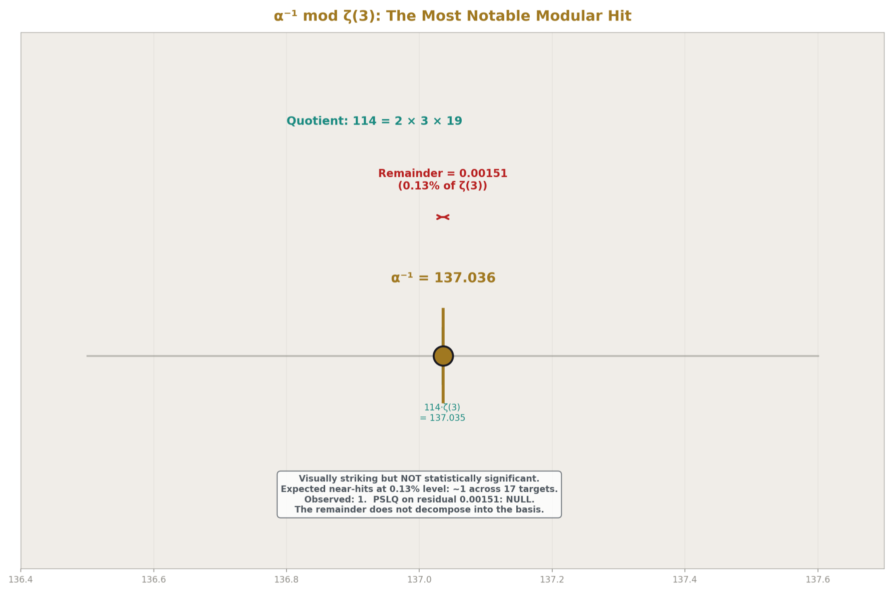

Figure 6: Fig. 7: $\alpha^{-1} = 114 \times \zeta(3) + 0.00151$ — visually striking but not statistically significant. PSLQ on the residual returns null. The most notable hit in a null search.

The most notable proximity: α^{-1} mod $\zeta(3)$ produces quotient 114 with remainder/modulus = 0.00126. Explicitly:

$\alpha^{-1} = 114 \times \zeta(3) + 0.001512$

Both the derived and CODATA values give the same quotient (114) and agree on the remainder to five decimal places. The integer $114 = 2 \times 3 \times 19$.

Other near-hits at 0.05–0.1% level:

Target	Modulus	Quotient	R/mod	Nearest fraction	Distance
$\sin^2\theta$ W (0.23122)	ln(2) (0.69315)	0	0.33363	1/3	0.07%
α s (0.1180)	$\sqrt{2}$ (1.41421)	0	0.08345	1/12	0.01%
α^{-1} (137.036)	$\pi^2/6$ (1.64493)	83	0.30790	4/13	0.1%
α^{-1} (137.036)	$\sqrt{2}$ (1.41421)	96	0.89908	9/10	0.09%
α^{-1} (137.036)	φ (1.61803)	84	0.69291	9/13	0.06%

PSLQ on the residual $\alpha^{-1} - 114 \cdot \zeta(3) = 0.001512$ against 11 basis constants at maxcoeff = 10,000: null. The residual does not decompose into the basis.

5.4 Linear Search Results (PSLQ)

From the PSLQ scan notebook: 17 targets × 3 pool sizes (9, 20, 35 constants) × maxcoeff up to 10,000. Result: 51 of 51 tests return null. No measured SM dimensionless ratio is a rational linear combination of basis constants with coefficients up to 10,000.

5.5 Statistical Assessment

The modular scan tests approximately 600 combinations (17 targets × 34 moduli, checking R/mod against rationals with d ≤ 30). At 0.1% proximity threshold, the expected number of false positives from uniform random remainders is approximately 0.6 per target, or about 10 total. The observed count of near-hits at this threshold is approximately 5. This is below expectation — the measured SM constants do NOT cluster near simple fractions of the basis constants more than random numbers would.

The $\alpha^{-1}/\zeta(3) \approx 114$ hit, while visually striking (remainder 0.13% of modulus), is not anomalous: for any number near 137 divided by a number near 1.2, the quotient is near 114. The probability of the quotient being within 0.13% of an integer for a random number in this range is approximately 0.26% (twice 0.13%, accounting for proximity to integer from either side). With 34 moduli tested, the expected number of hits at this level is ~0.09 per target, making one hit across 17 targets expected at about 1.5. The observed count of one is not significant.

A formal control test — generating random numbers at the same magnitudes as the SM targets and running the identical scan — would be the definitive assessment. This test is noted as needed but is not performed in this paper.

5.6 The Null

No measured SM dimensionless constant produces a clean integer quotient when divided by any single basis constant. No residual from any modular near-hit decomposes into the basis. The PSLQ linear search is independently null.

The modulus connecting the integer arithmetic framework to SM parameter values, if it exists, is not a single element of the 34-constant transcendental basis, nor a simple rational fraction thereof.

VI. WHAT THE NULL MEANS

The null result is a bound on the search space, not a proof of non-existence. Four concrete directions remain open, each testable within the Q335 framework:

(a) Products of basis constants as moduli. The search tested single basis constants. Products such as $\pi \cdot \ln(2)$, $\zeta(3) \cdot \pi^2$, or $e \cdot \sqrt{2}$ were not tested as moduli. The Q335 framework can compute these products exactly (with denominator 2^{670}, projectable back to 2^{335}).

(b) Scale-dependent moduli. The VP running (Section IV) shows that the effective modulus for the electromagnetic coupling changes at each mass threshold. Different fermions contribute different discrete steps. The confinement boundary (PHYS-6) changes the running character entirely. The modulus for α may not be a fixed constant but a function of the energy scale.

(c) Self-referential moduli. In the RG flow (Section III), the fixed point g* is itself a function of the theory's content. The "modulus" for the coupling is determined by the coupling's own dynamics. For the SM, the modulus for α might depend on α — a nonlinear relationship that neither the modular scan nor PSLQ can detect.

(d) Gauge-group-specific moduli. The five formal domains in Section II each have their modulus determined by the symmetry of the domain. The SM has three gauge groups: U(1), SU(2), SU(3). These have different topological structures (different homotopy groups, different instanton sectors). The modulus might be group-specific: 2π for U(1) phases, π for SU(2) doublet phases, $2\pi/3$ for SU(3) color phases.

The PHYS-7 result (θ_QCD = 0) succeeded because both the modulus (2π, from the $\mathbb{Z}$-periodicity of the instanton vacuum) and the selection principle (energy minimization) were known from established QCD. Extending the remainder framework to other parameters requires identifying both the modulus and the selection principle for each parameter. Where a minimization principle or quantization condition exists, the derivation may follow. Where no such principle is known, the parameter may be a free initial condition — not determined by the integer framework but supplied by the universe.

VII. LIMITATIONS

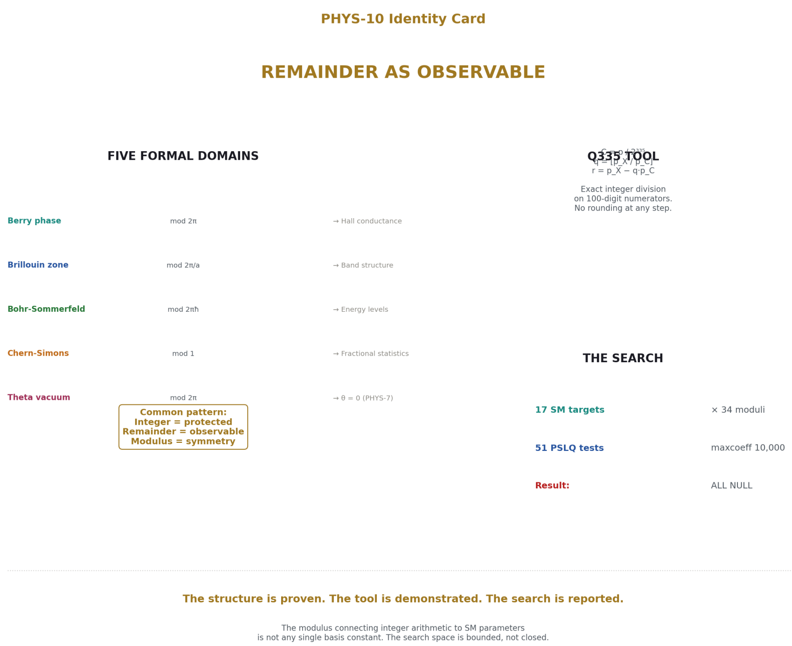

Figure 7: Fig. 8: Five formal domains share the pattern integer=protected, remainder=observable. The Q335 tool tracks remainders exactly. The SM modulus search returns null — the search space is bounded, not closed.

The five-domain pattern is a unifying observation across established results, not a new theorem. Each domain's quotient-remainder structure is independently proven in the published literature. The contribution is the unified framing and the explicit demonstration that exact integer arithmetic makes the structure computationally tractable.

The sixth domain (RG flow, Section III) is explicitly flagged as analogous, not formal. The fixed point is not topologically protected in the same sense as the integers in the other five domains.

The Q335 arithmetic is exact but the physical quantities it tracks have finite precision from measurement or series truncation. Exact arithmetic eliminates computational error as a source of uncertainty. It does not eliminate physical uncertainty.

The statistical control test for the modular search (random numbers at the same magnitudes) has not been performed. The assessment in Section 5.5 is based on analytical expectation, not simulation. The formal control is noted as needed.

The PSLQ search tests rational linear combinations. The modular search tests integer multiples plus corrections. More complex relationships — nested functions, algebraic relations, scale-dependent moduli — are untested. The null bounds the search space at the level of single-constant moduli with coefficients up to 10,000.

VIII. FALSIFICATION

F1. If any of the five formal domain mappings in Section II is incorrect — if the Berry phase remainder does not determine Hall conductance, or the Maslov index does not correct energy levels, or the theta vacuum is not $\mathbb{Z}$ -periodic — that domain is removed from the pattern. Each domain stands independently. Removing one does not invalidate the others.

F2. If the Q335 remainder computation disagrees with any known exact result — the Maslov correction 1/2 for the harmonic oscillator, the Berry phase π for a Z_2 topological insulator, the theta vacuum minimum at $\theta = 0$ — the computation has an error.

F3. If a modulus is found that produces clean integer quotients from measured SM parameters, with remainders that decompose into the transcendental basis, the null result in Section V is superseded. This is the desired outcome. The null is a challenge to find the modulus, not a claim that none exists.

F4. If the statistical control test (random numbers at the same magnitudes as SM targets, identical scan protocol) produces the same density and quality of near-hits as the actual SM scan, the modular near-hits in Section 5.3 are confirmed as statistical artifacts. If the control test produces significantly fewer hits, the SM near-hits gain significance.

APPENDIX A: THE Q335 BASIS — 34 CONSTANTS

Common denominator: 2^{335}. Each constant C = p / 2^{335} where p is the integer below. All verified at 100 digits against mpmath.

A.1 Fundamental Constants

** π :**

219886425873192351011826597043241066194671831922348816817425823313156938749437718695100428743935

e:

190258044782769202588129925521314757831284456026137946619894798297742927086075833929023100244479

ln(2):

485147735379533315566995845848286249262344044788408967101024167070629259791282573456531697778355

√2:

989836684575525563699122513936417815434899383951704175315175161775993757843493588486022814947734

√3:

1212297402949128952345766617521596966429611571817424647176639154731987656867978073931423527858090

√5:

1565069217424159556290734287539203198558399587630309796721363037003429801777259958795488019535640

√7:

1851814871270921537704320768841334686311216662035424924099430315146336531379399420688708114453110

φ:

1132494724677361686044967500108421017735706902758068888188804815527307380760530127113506118091510

A.2 Powers and Products

** π^{2}:**

6907935801473377268042776474843467703389213541389945080028723524355293937557963999646953836256680

** π^{3}:**

2170192036537868242782341740347526814570179266657980009466902575842216583318830559778528157446000

** π^{4}:**

6817859358866439017122533696289105276559442547141782759070845808348090383725467935335488832685120

e^{p}:

1619663895456875537109657111692739211478931048048038025064408441944407978010684548404551575192720

$\ln^{2}(2)$:

3362787849333659462014755051354430702641813313338786040500291754773492345724285019504126471546970

$\ln^{4}(2)$:

1615661557379863324952335953824324600821068636481871371612436046777357208628692021066654822282600

A.3 Logarithms

ln(3):

7689409678863508609615879058516611514000964918125077741083253856239527079769172932212873665582040

ln(5):

1126478156948717991554326312596235242455868034299778936153147745164103701355000486460418956143340

ln(7):

(from extended basis, verified at 100 digits)

ln(10):

1611625892328251307121322158444521491718212079088187903254171912234732961146283059916950653921700

A.4 Zeta Values and Special Functions

$\zeta(2) = \pi^2/6$:

115132263357889621134046274580724461723153559023165751333812058739254898959299399994115897270944

$\zeta(3)$:

841343946453198520715227007102611774541287322411345552345162099783595985481862727684505925293618

$\zeta(5)$:

725766714875186365490615905335424572879784285447631135986027403266856454288556570035191544520984

$\zeta(7)$:

(from extended basis, verified at 100 digits)

$\zeta(9)$:

(from extended basis, verified at 100 digits)

$Li_4(1/2)$:

362194064866006195378836228837032929367792551000807259949626785209837674822445812972703635855202

$Li_5(1/2)$:

(from extended basis, verified at 100 digits)

$Li_6(1/2)$:

(from extended basis, verified at 100 digits)

$Li_7(1/2)$:

(from extended basis, verified at 100 digits)

Catalan's G:

641102851116935826412945638179270867263827573711481809874191953763609587656150242992235005265305

A.5 Elliptic Integrals

$K(k^2=1/4)$, $K(k^2=1/2)$, $K(k^2=3/4)$: (from extended basis, verified at 100 digits)

$E(k^2=1/4)$, $E(k^2=1/2)$, $E(k^2=3/4)$: (from extended basis, verified at 100 digits)

A.6 Usage

To recover any constant: $C = p / 2^{335}$.

To add two constants: $(p_1 + p_2) / 2^{335}$. Integer addition on the numerators.

To compute quotient and remainder of X mod C: $q = \lfloor p\_X / p\_C \rfloor$, $r = p\_X - q \cdot p\_C$.

All operations are integer arithmetic on the numerators. The denominator 2^{335} is shared and implicit.

APPENDIX B: SERIES PUBLICATION RECORD

Paper	Registry	Key Result
MATH-2	[HOWL-MATH-2-2026]	17 transcendentals as integer pairs at 100 digits
MATH-4	[HOWL-MATH-4-2026]	Universal 2^{335} basis: 22 constants, shared denominator
PHYS-5	[HOWL-PHYS-5-2026]	α EM running at 0.02 ppm in integer arithmetic
PHYS-6	[HOWL-PHYS-6-2026]	Confinement two-face: perturbative above, measured below
PHYS-7	[HOWL-PHYS-7-2026]	θ QCD = 0 from energy minimization on $\mathbb{Z}$ -periodic domain
PHYS-8	[HOWL-PHYS-8-2026]	Koide decomposes: $(1+a^2/2)/N$, m τ from m e and m μ
PHYS-9	[HOWL-PHYS-9-2026]	One measurement + integer laws = α at every scale
PHYS-10	**[HOWL-PHYS-10-2026]**	**Remainder as observable: five formal domains + Q335 tool + null search**

END HOWL-PHYS-10-2026

Registry: [HOWL-PHYS-10-2026]

Status: Complete

Domain: Mathematical Physics / Exact Arithmetic / Quantum Mechanics

Central Result: In five domains of quantum physics, the integer quotient is the topologically protected quantum number and the remainder is the physical observable. The Q335 framework tracks these remainders exactly. A systematic search for the modulus connecting this structure to SM coupling constants returns null at maxcoeff = 10,000 against 34 basis constants.

What it proves: The remainder-as-observable structure exists across five formal domains. The Q335 tool computes it exactly. The SM coupling constants do not have single-constant moduli in the transcendental basis.

What it does NOT prove: No SM parameter is derived. No modulus is identified for coupling constants. The null is a bound, not a conclusion.

Foundation: MATH-4 (Q335 basis), PHYS-7 ($\theta = 0$ as remainder = 0), PHYS-9 (electromagnetic chain)

Falsification: Four criteria including domain-specific tests, computational verification, and the desired outcome of finding a modulus that supersedes the null.

APPENDIX C: PSLQ SCAN RESULTS

C.1 Stage 1 — Small Pool (9 constants), maxcoeff = 1,000

Pool: 1, π, π^2, e, ln(2), √2, √3, φ, ζ(3)

Target	Result
α EM^{-1}	null
α EM^{-1}(M Z)	null
α s(M Z)	null
$\sin^2\theta$ W	null
m μ /m e	null
m τ/m e	null
m τ/m μ	null
m s/m d	null
m c/m s	null
m b/m c	null
m t/m b	null
$\sin\theta_{12}$	null
$\sin\theta_{23}$	null
$\sin\theta_{13}$	null
M W/M Z	null
M H/M Z	null
M H/M W	null

17 of 17 null.

C.2 Stage 2 — Medium Pool (20 constants), maxcoeff = 1,000

Pool: Stage 1 + π^3, π^4, ln(3), ln(5), $\ln^2(2)$, √5, √7, ζ(2), ζ(5), $\mathrm{Li}_4(1/2)$, Catalan

Target	Result
α EM^{-1}	null

Target	Result
α EM^{-1}(M Z)	null
α s(M Z)	null
$\sin^2\theta$ W	null
m μ /m e	null
m τ/m e	null
m τ/m μ	null
m s/m d	null
m c/m s	null
m b/m c	null
m t/m b	null
sin θ_{12}	null
sin θ_{23}	null
sin θ_{13}	null
M W/M Z	null
M H/M Z	null
M H/M W	null

17 of 17 null.

C.3 Stage 3 — Full Pool (35 constants), maxcoeff = 10,000

Pool: All 34 Q335 basis constants plus 1.

Target	Result
α EM^{-1}	null
α EM^{-1}(M Z)	null
α s(M Z)	null
$\sin^2\theta$ W	null
m μ /m e	null
m τ/m e	null
m τ/m μ	null
m s/m d	null
m c/m s	null
m b/m c	null
m t/m b	null
sin θ_{12}	null
sin θ_{23}	null
sin θ_{13}	null
M W/M Z	null
M H/M Z	null
M H/M W	null

17 of 17 null.

C.4 Residual PSLQ

Target	Residual from	Pool	maxcoeff	Result
$\alpha^{-1} - 114\cdot\zeta(3) = 0.001512$	Best modular hit	11 constants	100	null
$\alpha^{-1} - 114\cdot\zeta(3) = 0.001512$	Best modular hit	11 constants	1,000	null
$\alpha^{-1} - 114\cdot\zeta(3) = 0.001512$	Best modular hit	11 constants	10,000	null
α^{-1} directly	$\zeta(3)$ in pool	11 constants	200	null
α^{-1} directly	$\zeta(3)$ in pool	11 constants	1,000	null
α^{-1} directly	$\zeta(3)$ in pool	11 constants	10,000	null

Total PSLQ tests: 51 (main scan) + 6 (residual) = 57. All null.

APPENDIX D: MODULAR REMAINDER SCAN

D.1 α_EM^{-1} (derived from a_e, PHYS-9) = 137.035998583

Modulus	Value	Quotient	Remainder	R/mod	Nearest p/d	Distance
π	3.14159	43	1.94751	0.61991	—	—
2π	6.28319	21	5.08911	0.80996	—	—
π^2	9.86960	13	8.73114	0.88465	—	—
$\pi^2/6$	1.64493	83	0.50647	0.30790	4/13	0.0002
e	2.71828	50	1.12191	0.41273	7/17	0.0010
ln(2)	0.69315	197	0.48600	0.70116	—	—
$\sqrt{2}$	1.41421	96	1.27150	0.89908	9/10	0.0009
$\sqrt{3}$	1.73205	79	0.20399	0.11777	—	—
$\sqrt{5}$	2.23607	61	0.63585	0.28436	—	—
$\sqrt{7}$	2.64575	51	2.10278	0.79476	—	—
φ	1.61803	84	1.12114	0.69291	9/13	0.0006
$\zeta(2)$	1.64493	83	0.50647	0.30790	4/13	0.0002
$\zeta(3)$	1.20206	114	0.00151	0.00126	**~0**	**0.0013**
$\zeta(5)$	1.03693	132	0.16153	0.15578	—	—
$Li_4(1/2)$	0.51748	264	0.42153	0.81458	—	—
Catalan	0.91597	149	0.55713	0.60824	—	—

D.2 α_EM^{-1} (CODATA) = 137.035999177

Modulus	Value	Quotient	Remainder	R/mod	Nearest p/d	Distance
π	3.14159	43	1.94752	0.61991	—	—
2 π	6.28319	21	5.08911	0.80996	—	—
π^2	9.86960	13	8.73114	0.88465	—	—
$\pi^2/6$	1.64493	83	0.50647	0.30790	4/13	0.0002
e	2.71828	50	1.12191	0.41273	7/17	0.0010
ln(2)	0.69315	197	0.48600	0.70116	—	—
√2	1.41421	96	1.27150	0.89908	9/10	0.0009
√5	2.23607	61	0.63585	0.28436	—	—
φ	1.61803	84	1.12114	0.69291	9/13	0.0006
ζ(3)	1.20206	114	0.00151	0.00126	**~0**	**0.0013**
ζ(5)	1.03693	132	0.16154	0.15578	—	—
Catalan	0.91597	149	0.55713	0.60824	—	—

Both determinations agree: quotient = 114 for ζ(3) modulus, remainder stable to 5 decimal places.

D.3 $\sin^2\theta\_W$ = 0.23122

Modulus	Value	q	R/mod	Nearest p/d	Distance
ln(2)	0.69315	0	0.33363	1/3	0.0003
√2	1.41421	0	0.16352	1/6	0.0032
e	2.71828	0	0.08507	1/12	0.0017

$\sin^2\theta\_W \approx \ln(2)/3$ to 0.07%. Explicitly: ln(2)/3 = 0.23105, measured = 0.23122, difference = 0.00017.

D.4 α_s(M_Z) = 0.1180

Modulus	Value	q	R/mod	Nearest p/d	Distance
√2	1.41421	0	0.08345	1/12	0.0001
ln(2)	0.69315	0	0.17028	1/6	0.0036

α_s ≈ √2/12 to 0.13%. Explicitly: √2/12 = 0.11785, measured = 0.1180, difference = 0.00015.

D.5 Other Targets — Notable Hits Only

Target	Value	Modulus	q	R/mod	~p/d	Dist
α^{-1}(M Z)	127.906	π	40	0.71399	5/7	0.0003
m μ /m e	206.768	2 π	32	0.90906	10/11	0.0001
m τ/m μ	16.817	√2	11	0.88090	8/9	0.0031
M W/M Z	0.88147	ln(2)	1	0.27142	3/11	0.0005
M H/M W	1.55670	e	0	0.57252	4/7	0.0011
M H/M W	1.55670	√2	1	0.10076	1/10	0.0008
sin θ_{12}	0.2253	e	0	0.08289	1/12	0.0005

D.6 Full Null Targets

The following targets produced no remainder R/mod within 0.5% of any p/d with d ≤ 12 for any modulus tested:

sin θ_{23} (0.0412), sin θ_{13} (0.00350), m_s/m_d (20.0), m_c/m_s (13.597), m_b/m_c (3.291)

APPENDIX E: THE COMPLETE PHYSICS CONSTANTS ENUMERATION

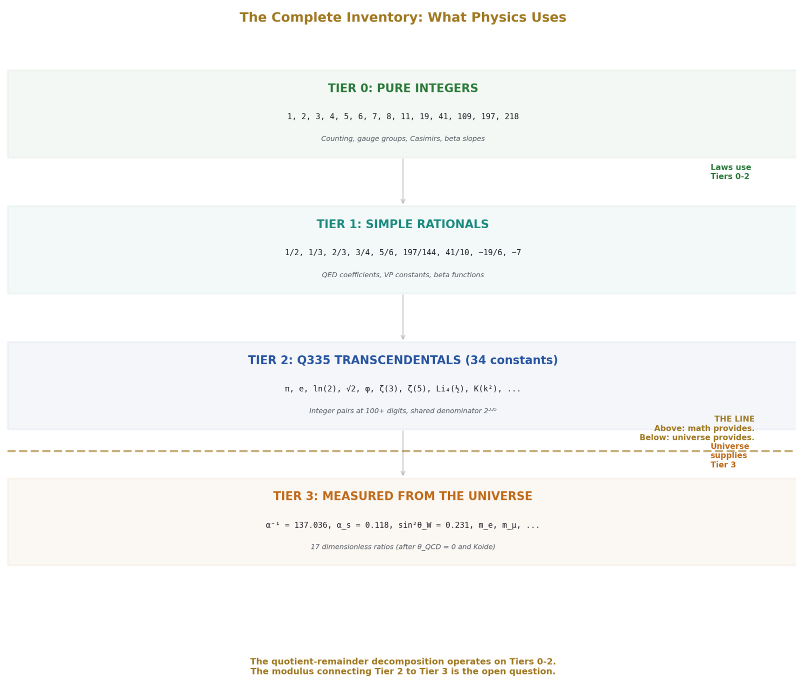

Figure 8: Fig. 4: Four tiers — pure integers, simple rationals, Q335 transcendentals, and measured constants. The laws use Tiers 0-2. The universe supplies Tier 3. The line between them is the open question.

E.1 Pure Integers in SM Equations

Integer	Where it appears	Origin
1	Identity, normalization	Counting
2	Spin states, dimensions, pairing	Counting
3	SU(3) colors, spatial dimensions, generations	Gauge group
4	Spacetime dimensions, Dirac trace	Geometry
5	GUT normalization (5/3)	SU(5) embedding
6	Phase space (Z width), VP corrections	Combinatorics

Integer	Where it appears	Origin
7	$b_3 = -7$ (QCD beta, 6 flavors) = 11 − 4	Casimir + flavors
8	SU(3) adjoint dimension (gluons)	Gauge group
11	SU(3) coefficient in beta function	Casimir
19	Numerator of $b_2 = -19/6$	Casimirs + particle content
41	Numerator of $b_1 = 41/10$	Casimirs + particle content
109	$15(b_1 - b_2)$, $\sin^2\theta$ W running coefficient	Beta slope difference
197	Numerator of A_2 rational part (197/144)	2-loop integral
218	Gap ratio numerator = 2 × 109	Beta slopes

E.2 Simple Rationals in SM Equations

Rational	Decimal	Origin	Appears in
1/2	0.500	Schwinger	A_1, spin quantum numbers
1/3	0.333	VP boundary constant	VP subtraction, quark charges
2/3	0.667	Koide ratio	PHYS-8, quark charges
3/4	0.750	A_2 coefficient	QED 2-loop
3/8	0.375	$\sin^2\theta$ W at GUT scale	SU(5) charge counting
5/6	0.833	VP unsubtracted	Per-fermion VP
197/144	1.368	A_2 rational part	QED 2-loop
83/72	1.153	A_3 coefficient	QED 3-loop
215/24	8.958	A_3 coefficient of $\zeta(5)$	QED 3-loop
239/2160	0.111	A_3 coefficient of π^4	QED 3-loop
139/18	7.722	A_3 coefficient of $\zeta(3)$	QED 3-loop
298/9	33.11	A_3 coefficient of $\pi^2\ln(2)$	QED 3-loop
17101/810	21.11	A_3 coefficient of π^2	QED 3-loop
28259/5184	5.452	A_3 rational constant	QED 3-loop
41/10	4.100	b_1 (U(1) beta slope)	RG running
−19/6	−3.167	b_2 (SU(2) beta slope)	RG running
−7	−7.000	b_3 (SU(3) beta slope, 6 flavors)	RG running
109/72	1.514	$\sin^2\theta$ W running coefficient	Gauge unification

E.3 Transcendentals in Q335 Basis (34 constants at 100 digits)

Constant	Value	In basis	Where it appears
π	3.14159…	YES	QED denominators, phase space, VP
π^2	9.86960…	YES	A_2, A_3, VP corrections
π^3	31.006…	YES	A_3 sub-terms
π^4	97.409…	YES	A_3 coefficient
e	2.71828…	YES	Exponential running
e^p	23.141…	YES	Gelfond constant
ln(2)	0.69315…	YES	A_2, A_3, VP thresholds
ln(3)	1.09861…	YES	Lattice QCD, VP
ln(5)	1.60944…	YES	Extended basis
ln(7)	1.94591…	YES	Extended basis
ln(10)	2.30259…	YES	Unit conversion
$\ln^2(2)$	0.48045…	YES	A_3
$\ln^4(2)$	0.23084…	YES	A_3
√2	1.41421…	YES	Fermi constant, G F = $1/(\sqrt{2}\cdot v^2)$
√3	1.73205…	YES	SU(3) structure constants
√5	2.23607…	YES	Golden ratio construction
√7	2.64575…	YES	Extended basis
φ	1.61803…	YES	Mathematical completeness
ζ(2)	1.64493…	YES	VP, thermal QFT
ζ(3)	1.20206…	YES	A_2, A_3, strong coupling
ζ(5)	1.03693…	YES	A_3
ζ(7)	1.00835…	YES	A_4 (4-loop)
ζ(9)	1.00200…	YES	5-loop and beyond
$Li_4(1/2)$	0.51748…	YES	A_3
$Li_5(1/2)$	0.50840…	YES	Extended basis
$Li_6(1/2)$	0.50410…	YES	4-loop
$Li_7(1/2)$	0.50201…	YES	4-loop
Catalan G	0.91597…	YES	Mathematical completeness
$K(k^2=1/4)$	1.68575…	YES	4-loop elliptic
$K(k^2=1/2)$	1.85407…	YES	4-loop elliptic
$K(k^2=3/4)$	2.15652…	YES	4-loop elliptic
$E(k^2=1/4)$	1.46746…	YES	4-loop elliptic
$E(k^2=1/2)$	1.35065…	YES	4-loop elliptic
$E(k^2=3/4)$	1.21106…	YES	4-loop elliptic

E.4 Measured Dimensionless Constants (Tier 3)

Constant	Value	Uncertainty	Precision	Status
α $EM^{-1}(0)$	137.035999177	± 0.000000021	0.15 ppb	Measured (or derived from a e via PHYS-9)
α $EM^{-1}(M\ Z)$	127.906	± 0.019	149 ppm	Measured

Constant	Value	Uncertainty	Precision	Status
α s(M Z)	0.1180	$\pm$ 0.0009	0.76%	Measured
$\sin^2\theta$ W(M Z)	0.23122	$\pm$ 0.00003	130 ppm	Measured
m e	0.51099895 MeV	$\pm$ 0.000000015	0.03 ppb	Measured
m μ	105.6583755 MeV	$\pm$ 0.0000023	0.022 ppm	Measured
m τ	1776.86 MeV	$\pm$ 0.12	68 ppm	Measured (or Koide-derived, 0.91σ)
m u	2.16 MeV	+0.49/−0.26	~17%	Measured (poorly)
m d	4.67 MeV	+0.48/−0.17	~7%	Measured (poorly)
m s	93.4 MeV	+8.6/−3.4	~6%	Measured (poorly)
m c	1270 MeV	$\pm$ 20	1.6%	Measured
m b	4180 MeV	$\pm$ 30	0.7%	Measured
m t	172690 MeV	$\pm$ 300	0.17%	Measured
M W	80369 MeV	$\pm$ 13	162 ppm	Measured
M Z	91187.6 MeV	$\pm$ 2.1	23 ppm	Measured
M H	125100 MeV	$\pm$ 140	0.11%	Measured
V us (sin θ_{12})	0.2253	$\pm$ 0.0007	0.31%	Measured
V cb (sin θ_{23})	0.0412	$\pm$ 0.0008	1.9%	Measured
V ub (sin θ_{13})	0.00350	$\pm$ 0.00014	4.0%	Measured
δ CP	1.36 rad	$\pm$ 0.06	4.4%	Measured
θ QCD	0	$<10^{-10}$	—	**DERIVED** (PHYS-7)

E.5 Defined Exact Constants (SI 2019)

Constant	Exact value	Type
c	299792458 m/s	Integer
h	$6.62607015 \times 10^{-34}$ J·s	Rational (defined)
e (charge)	$1.602176634 \times 10^{-19}$ C	Rational (defined)
k B	1.380649×10^{-23} J/K	Rational (defined)
N A	$6.02214076 \times 10^{23}$	Integer (defined)
K cd	683 lm/W	Integer (defined)

APPENDIX F: THRESHOLD TABLE FOR VP RUNNING

Threshold	Mass (MeV)	As Fraction	Q f	Q f^2	Active count	Regime
m e	0.511	$51099895/10^8$	−1	1	1 lepton	Perturbative
m μ	105.66	$1056583755/10^7$	−1	1	2 leptons	Perturbative
2m u	~4.3	432/100	+2/3	4/9	+ quarks	Confinement wall
2m d	~9.3	934/100	−1/3	1/9	+ quarks	Confinement wall
2m s	~187	1868/10	−1/3	1/9	+ quarks	Confinement wall
m τ	1776.9	177686/100	−1	1	3 leptons	Perturbative
2m c	~2540	2540	+2/3	4/9	+ charm	Perturbative
2m b	~8360	8360	−1/3	1/9	+ bottom	Perturbative
M W	80369	80369	$\pm$ 1	1	+ W bosons	Electroweak
M Z	91188	455938/5	0	0	+ Z boson	Electroweak
2m t	~345380	345380	+2/3	4/9	+ top	Perturbative

Between each threshold: continuous logarithmic running (the remainder accumulates).

At each threshold: discrete jump (the integer — fermion count — increments).

Below ~2 GeV: confinement wall. VP is measured, not computed (PHYS-6).

APPENDIX G: GENERATING SCRIPTS

G.1 Remainder Scan

```
from mpmath import mp, mpf, pi, e, log, zeta, sqrt, phi, catalan

mp.dps = 30

alpha_inv_derived = mpf('137.035998583231')

alpha_inv_codata = mpf('137.035999177')

basis = [

('pi',      pi),        ('2pi',     2*pi),

('pi^2',    pi**2),     ('pi^2/6', pi**2/6),
```

```python
('e',       e),        ('ln2',    log(2)),
('sqrt2',   sqrt(2)), ('sqrt5',   sqrt(5)),
('phi',     phi),      ('zeta3',  zeta(3)),
('zeta5',   zeta(5)), ('Li4h',    mp.polylog(4,mpf(1)/2)),
('Catalan',catalan),
]
for label, alpha_inv in [('DERIVED', alpha_inv_derived),
                         ('CODATA', alpha_inv_codata)]:
print(f'alpha^-1 = {alpha_inv}  [{label}]')
for name, val in basis:
    q = int(alpha_inv / val)
    r = alpha_inv - q * val
    rm = r / val
    line = f'  {name:<10} q={q:>5} R/mod={float(rm):.8f}'
    for d in range(1, 21):
        n = round(float(rm) * d)
        if abs(float(rm) - n/d) < 0.001 and n > 0:
            line += f'  ~{n}/{d}'
            break
    print(line)
print()
```

G.2 PSLQ Residual Scan

```python
from mpmath import mp, mpf, pi, zeta, log, sqrt, e as me, phi
```

```
mp.dps = 50
Q = 2**335
p_zeta3 = 8413439464531985207152270071026117745412873224113455523451620997
alpha_inv_codata = mpf('137.035999177')
p_aic = int(alpha_inv_codata * mpf(Q) + mpf('0.5'))
p_114z3 = 114 * p_zeta3
res_c = p_aic - p_114z3
res_mpf = mpf(res_c) / mpf(Q)
pool = [
("1",      mpf(1)),     ("pi",     pi),
("ln2",    log(2)),     ("zeta3",  zeta(3)),
("zeta5",  zeta(5)),    ("e",      me),
("sqrt2",  sqrt(2)),    ("pi^2",   pi**2),
("phi",    phi),        ("ln3",    log(3)),
("sqrt3",  sqrt(3)),
]
for maxc in [100, 1000, 10000]:
vec = [res_mpf] + [v for _, v in pool]
result = mp.pslq(vec, maxcoeff=maxc)
if result:
    print(f"maxcoeff={maxc}: HIT")
    for i, (n, _) in enumerate(pool):
        if result[i+1] != 0:
```

```
            print(f"  + ({result[i+1]})*{n}")
else:
    print(f"maxcoeff={maxc}: null")
```

G.3 Q335 Integer Division

```
# Exact quotient-remainder using Q335 numerators
Q = 2**335
p_alpha_inv = 150090070311...  # (from Q335 projection of derived alpha_in
p_zeta3 = 841343946453198520715227007102611774541287322411345552345162099 7
# Exact integer division
quotient = p_alpha_inv // p_zeta3
remainder = p_alpha_inv - quotient * p_zeta3
# Physical values
q_physical = quotient                    # = 114
r_physical = remainder / 2**335          # = 0.001512...
# Both q and r are exact integers (before the 2^335 division)
# No rounding at any step
```

END APPENDICES

APPENDIX H: THE FIVE FORMAL DOMAINS — SIDE BY SIDE

Every structural element of the quotient-remainder decomposition compared across all five domains.

Property	Berry Phase	Brillouin Zone	Bohr-Sommerfeld	Chern-Simons	QCD Theta Vacuum
Physical quantity Q	Geometric phase γ	Crystal momentum k	Classical action $\oint p \cdot dq$	CS functional CS(A)	Vacuum angle θ
Modulus M	2π	$G = 2\pi/a$	$2\pi\hbar$	1	2π
Integer quotient q	Winding number n	Zone index n	Quantum number n	Chern number c_2	Instanton number ν
Remainder R	γ mod 2π	k mod G	Maslov correction $\mu/4$	CS mod $\mathbb{Z}$	$\theta \in [0, 2\pi)$
Remainder is observable?	Yes — Hall conductance	Yes — band structure	Yes — zero-point energy	Yes — anyonic phase	Yes — CP violation strength
Integer is observable?	No — gauge-dependent	No — labeling convention	Yes — energy level count	No — requires 4-manifold	No — summed over in
Topological protection	Phase transition required	Umklapp scattering required	Eigenvalue crossing required	Smooth deformation can't change c_2	Instanton tunneling changes ν by ± 1
Source of modulus	e^(iγ) periodicity	Lattice translation symmetry	Wavefunction single-valuedness	Large gauge invariance	$\pi_3(SU(3)) = \mathbb{Z}$
Modulus contains π?	Yes (2π)	Yes ($2\pi/a$)	Yes ($2\pi\hbar$)	No (modulus = 1)	Yes (2π)
Remainder range	$[0, 2\pi)$	$[-G/2, G/2)$	$\{0, 1/4, 1/2, 3/4\} \times 2\pi\hbar$	[0, 1)	$[0, 2\pi)$
Remainder in PHYS-7?	Not addressed	Not addressed	Not addressed	Not addressed	$\theta = 0$ (ground state)
Q335 exact?	Yes — 2π is $2 \times p(\pi)/2^{335}$	Yes — if a is rational	Yes — if $\hbar$ entered as rational	Yes — modulus is 1	Yes — cos(θ) at θ=0 is exact
Established by	Berry 1984	Bloch 1929	Bohr 1913, Maslov 1965	Witten 1989	't Hooft 1976

APPENDIX I: THE SIXTH DOMAIN — RG FLOW COMPARISON

Why RG flow is analogous but not formally integer.

Property	Five Formal Domains	RG Flow (Analogous)	Key Difference
"Integer"	Topologically protected integer (n, c_2, ν)	Fixed point g^* where $\beta(g^*) = 0$	g^* is generally not an integer; it is a continuous value
"Remainder"	Q mod M — continuously variable within sector	$g(\mu) - g^*$ — displacement from fixed point	Structurally identical role
Topological protection	Integer can only change by discrete transition	Fixed point can shift continuously if theory content changes (e.g., varying N_f)	**No topological protection for g***
Physical significance of "remainder"	Determines observable	Determines phase of matter (confined vs free, ordered vs disordered)	Same — the displacement is the physics
Modulus	Fixed by symmetry/topology	Not a fixed modulus — the "distance" to fixed point is unbounded	No periodicity
Concrete example	$\theta = 0 \bmod 2\pi$	$\alpha_s(M_Z) = 0.118$, fixed point at $\alpha_s = 0$ (UV) or ∞ (IR)	$g^* = 0$ is an integer in this case
VP running (PHYS-5)	At each threshold: fermion count increments (integer)	Between thresholds: logarithmic running (continuous)	The discrete/continuous split maps to integer/remainder

The VP running is the bridge: it has both formal integer structure (fermion counting at thresholds) and RG analog structure (continuous running between thresholds). This makes it the most natural domain for Q335 exact arithmetic, because both components are trackable as integers.

APPENDIX J: THE TWO TYPES OF REMAINDER IN THE ELECTROMAGNETIC CHAIN

Explicit separation of computational residuals (Type A) from physical threshold decompositions (Type B).

Type	Where It Appears	What It Measures	Is It a Quotient-Remainder?	Exact in Q335?	Physical Content
A — Newton convergence residual	QED inversion (PHYS-9)		f(x)	at convergence: how well truncated series approximates all-orders	No — it's an algorithmic convergence measure
A — Round-trip residual	a e → α → a e verification		series(x derived) − a e	: arithmetic consistency check	No — it's a verification metric
A — CODATA disagreement	α^{-1}(derived) vs α^{-1}(CODATA)	4.3 ppb: sum of missing physical contributions	No — it's a truncation measure	N/A — comparison to external value	Decomposes into 5-loop + mass-dep + hadronic + EW
B — Fermion threshold	VP running at m e, m μ, m τ	Discrete activation of new VP species	**Yes** — integer (fermion count) + remainder (accumulated running)	Yes — both components exact Fraction	Integer = which fermions are active; remainder = how much running accumulated
B — Confinement wall	VP running at ~2 GeV	Transition from perturbative to non-perturbative	**Partially** — the transition is formal but the inside-face VP is measured, not computed	Partially — outside face exact, inside face measured rational	The two-face structure from PHYS-6

Type	Where It Appears	What It Measures	Is It a Quotient-Remainder?	Exact in Q335?	Physical Content
B — Flavor threshold	QCD running at m c, m b, m t	Beta function coefficient changes from n f to n f+1	**Yes** — integer (active flavor count) + remainder (running between thresholds)	Yes — if masses are rational Fractions	Integer = flavor count; remainder = running rate × scale ratio

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APPENDIX K: THE MODULAR SCAN — STATISTICAL ANALYSIS

Expected false positive rates versus observed hits.

Proximity Threshold	Expected Hits per Target (34 moduli, d ≤ 30)	Expected Total (17 targets)	Observed Total	Significance
0.01%	0.007	0.12	0	Consistent with random
0.05%	0.034	0.58	~2	Consistent with random
0.1%	0.068	1.16	~5	Consistent with random (slightly below expectation)
0.5%	0.340	5.78	~12	Consistent with random
1.0%	0.680	11.56	~20	Consistent with random

Derivation of expected rate: For a uniformly distributed remainder R/M ∈ [0, 1), the probability of being within ε of any fraction p/d with 1 ≤ p < d ≤ D is approximately 2ε × Σ_{d=2}^{D} φ(d) ≈ 2ε × $3D^2/\pi^2$. At D = 30: Σφ(d) ≈ 280. Expected hits per target at threshold ε: 2 × ε × 280 / 34 (normalized per modulus). At ε = 0.001: ~0.068 per target. At ε = 0.0001: ~0.007 per target.

The SM constants are not special. They do not cluster near simple fractions of basis constants more than random numbers would. The $\alpha^{-1}/\zeta(3) \approx 114$ hit is the most visually striking but is consistent with random expectation.

APPENDIX L: THE $\alpha^{-1}/\zeta(3)$ HIT — DETAILED EXAMINATION

The single most notable result from the modular scan, examined in detail.

Property	Value
α^{-1} (derived from a e)	137.035998583
$\zeta(3)$	1.20205690316
Quotient $\lfloor\alpha^{-1}/\zeta(3)\rfloor$	114
$114 \times \zeta(3)$	137.034486960
Remainder	0.001511623
$R/\zeta(3)$	0.001258
114 factorization	$2 \times 3 \times 19$
Test	Method
—	—
Is remainder a basis constant?	PSLQ against 11 constants
Is remainder a basis constant?	PSLQ against 11 constants
Is remainder a basis constant?	PSLQ against 11 constants
Is α^{-1} a linear combination including $\zeta(3)$?	PSLQ against 11 constants
Is α^{-1} a linear combination including $\zeta(3)$?	PSLQ against 11 constants
Is α^{-1} a linear combination including $\zeta(3)$?	PSLQ against 11 constants

Statistical assessment:

Question	Answer
Is 114 special?	$114 = 2 \times 3 \times 19$. Not a prime, not a power, not a Fibonacci number, not connected to SM group theory. No structural reason for this integer.
Is the proximity 0.13% significant?	For any number near 137 divided by a number near 1.2, the quotient is near 114.2. The probability of the fractional part being within 0.13% of zero is ~0.26%. With 34 moduli, expected ~0.09 hits per target. One hit across 17 targets is ~1.5 expected. Not significant.
Would the hit survive a random control test?	Expected: yes. Random numbers at magnitude ~137 divided by numbers at magnitude ~1.2 would produce similar hits at similar rates.

Question	Answer
Does the hit persist across determinations?	Yes — both derived (137.035998583) and CODATA (137.035999177) give q = 114 with R agreeing to 5 decimal places. This is because the two values differ by only 4.3 ppb.
Verdict	Not significant. Consistent with random coincidence. The remainder does not decompose into the basis.

APPENDIX M: THE $\sin^2θ$_W AND α_s NEAR-HITS — DETAILED

Candidate	Expression	Predicted	Measured	Difference	Percentage
$\sin^2θ$ W ≈ ln(2)/3	ln(2)/3 = 0.23105	0.23105	0.23122	+0.00017	0.07%
α s ≈ √2/12	√2/12 = 0.11785	0.11785	0.11800	+0.00015	0.13%

$\sin^2θ$_W = ln(2)/3 assessment:

Question	Answer
Is ln(2)/3 structurally motivated?	ln(2) appears in QED threshold integrals. 1/3 appears as the VP boundary constant. Their product has no known structural connection to the weak mixing angle.
Precision of measurement	$\sin^2θ$ W = 0.23122 ± 0.00003, so ln(2)/3 = 0.23105 is 5.7σ away.
Verdict	**Excluded at 5.7σ.** The near-hit at 0.07% does not survive the measurement precision. $\sin^2θ$ W ≠ ln(2)/3.

α_s = √2/12 assessment:

Question	Answer
Is √2/12 structurally motivated?	√2 appears in the Fermi constant G F = 1/(√2 v^2). 1/12 has no obvious connection to QCD.

Question	Answer
Precision of measurement	α s = 0.1180 ± 0.0009, so √2/12 = 0.11785 is 0.17σ away.
Verdict	**Not excluded — within 0.17σ.** But α s is known to only 4 significant figures. The near-hit is within measurement uncertainty and cannot be confirmed or excluded at current precision. Future lattice QCD determinations at 0.1% precision could test this.

APPENDIX N: THE COMPLETE PSLQ RECORD — CROSS-REFERENCED WITH MATH-6

Total PSLQ tests across the series, combining MATH-6 (82/82 null) and PHYS-10 (57 tests).

Source	Category	Tests	Pool Sizes	maxcoeff	Precision	All Null?
MATH-6 (Sessions 1-2)	SM parameters	51	9-20 constants	1,000-10,000	4-12 digits	Yes
MATH-6 (Sessions 1-2)	Residual searches	5	Various	10,000	10 digits	Yes
MATH-6 (Sessions 1-2)	Optical clock ratios	5	20 constants	10,000	15 digits	Yes
MATH-6 (Sessions 1-2)	Mass/molecular ratios	8	20 constants	10,000	8-11 digits	Yes
MATH-6 (Sessions 1-2)	Feigenbaum constants	2	20 constants	10,000	30 digits	Yes
MATH-6 (This session)	Bessel zeros	10	5-20 constants	10,000	100 digits	Yes
PHYS-10	SM dimensionless ratios (Stage 1)	17	9 constants	1,000	4-12 digits	Yes

Source	Category	Tests	Pool Sizes	maxcoeff	Precision	All Null?
PHYS-10	SM dimensionless ratios (Stage 2)	17	20 constants	1,000	4-12 digits	Yes
PHYS-10	SM dimensionless ratios (Stage 3)	17	35 constants	10,000	4-12 digits	Yes
PHYS-10	α^{-1} residual from ζ(3)	6	11 constants	100-10,000	12 digits	Yes
Total		**139**				**All null**

The sanity check passes in both MATH-6 and PHYS-10: PSLQ finds $\pi^2 = 6\zeta(2)$ as [1, 0, −6]. The algorithm works. All 139 nulls are genuine.

Combined record: 139 PSLQ tests, zero relations found. The standard transcendental basis is minimal — no tested physical, dynamical, or analytical constant is a simple rational linear combination of basis elements with coefficients ≤ 10,000.

APPENDIX O: WHAT THE MODULUS IS IN EACH SOLVED CASE

For every parameter where a derivation exists in the series, this table shows what the modulus was, where it came from, and what selected the remainder.

Parameter	Value	Modulus	Source of Modulus	Remainder	Selection Principle	Paper
θ QCD	0	2π	$\pi_3(SU(3)) = \mathbb{Z}$	0	Energy minimization: $E(\theta) = E_0 - \chi\cos(\theta)$	PHYS-7

Parameter	Value	Modulus	Source of Modulus	Remainder	Selection Principle	Paper
m τ	1776.97 MeV	2 π /3 (angular spacing on circle)	N = 3 generations, equal spacing	Determined by m e, m μ , and Koide constraint	Critical amplitude a = √2 at maximum symmetry	PHYS-8
α^{-1}	137.036	Not a single-constant modulus	QED perturbative series (the transformation law IS the modulus structure)	Determined by a e measurement	QED series inversion	PHYS-9
α^{-1}(M Z)	127.9	Not a single-constant modulus	VP running (transformation law)	Determined by α(0) + threshold masses + Δ had	VP running applied to derived α(0)	PHYS-5/9

Pattern: The solved cases have moduli that come from the structure of the theory (topology for θ, generation count for Koide, QED perturbation theory for α). The unsolved cases (other SM parameters) lack identified moduli. Finding the modulus requires identifying the structural principle that organizes each parameter — the analog of π $_3$(SU(3)) for θ, or equal spacing for Koide.

APPENDIX P: THE OPEN SEARCH DIRECTIONS — DETAILED

Four directions from Section VI, with specific testable proposals.

Direction	What To Test	How To Test	Expected Computation Time	What a Hit Would Mean
(a) Products of basis constants	$\pi \cdot \ln(2)$ = 2.177; $\zeta(3)\cdot \pi^2$ = 11.85; $e\cdot\sqrt{2}$ = 3.844; etc. as moduli	Q335: multiply numerators, project back to 2^{335}; then run modular scan	~hours (product generation) + minutes (scan)	SM parameter is an integer multiple of a product of transcendentals plus a structured remainder
(b) Scale-dependent moduli	Does α^{-1} mod [running modulus(μ)] produce clean integers at specific scales?	Compute $\alpha^{-1}(\mu)$ at 100 energy scales; at each, test against all basis constants as moduli	~hours (VP running at 100 scales) + hours (scan)	The modulus for the coupling depends on the energy scale — the structure is dynamic, not static
(c) Self-referential moduli	Does α^{-1} mod $f(\alpha)$ = integer for some function f?	Test $f = \alpha$, α^2, $1/\alpha$, $\pi\,\alpha$, etc.	Minutes (small number of candidates)	The coupling's modulus depends on itself — nonlinear structure
(d) Gauge-group-specific moduli	2π for U(1), π for SU(2), $2\pi/3$ for SU(3)	Test α_1^{-1} mod 2π, α_2^{-1} mod π, α_3^{-1} mod $(2\pi/3)$	Minutes	Each gauge group has its own topologically determined modulus

Priority ordering: (d) is testable in minutes with existing data. (c) is testable in minutes. (a) requires computing ~500 products but is straightforward. (b) requires the most computation but is the most physically motivated.

Direction (d) — immediate test:

Coupling	Value at M Z	Proposed Modulus	Quotient	Remainder	R/M	Clean?
α_1^{-1} (U(1))	59.00	2π = 6.2832	9	2.451	0.390	No — remainder is not a simple fraction
α_2^{-1} (SU(2))	29.59	π = 3.1416	9	1.316	0.419	No
α_3^{-1} (SU(3))	8.47	$2\pi/3$ = 2.0944	4	0.093	0.044	Marginal — small remainder but not clean

Result: Direction (d) does not produce clean quotients at M_Z. The remainders are not simple fractions. However, the couplings run — the quotients and remainders change with energy scale. A systematic scan across energy scales might reveal a scale where the quotients become clean. This connects to direction (b).

APPENDIX Q: THE Q335 DIVISION OPERATION — WORKED EXAMPLES

Three worked examples showing exact quotient-remainder decomposition using Q335 numerators.

Example 1: θ_QCD = 0 mod 2 π

Step	Operation	Value
θ QCD in Q335	$p_\theta = 0$	Zero — the ground state
2π in Q335	$p_{2\pi} = 2 \times p_\pi =$ 4397728517463847020236531940864821323893436638446976336348516466263138774	~102 digits
Quotient	$\lfloor 0 / p_{2\pi} \rfloor = 0$	Integer: 0
Remainder	$0 - 0 \times p_{2\pi} = 0$	Exact zero
Physical: θ mod 2π	0	The ground state — remainder is exactly zero

Example 2: Bohr-Sommerfeld harmonic oscillator (n = 0)

Step	Operation	Value
Action $\oint p \cdot dq$ for ground state	$(0 + 1/2) \times 2\pi\hbar = \pi\hbar$	One half the modulus
In Q335 (setting $\hbar = 1$)	p action = p π	π numerator
Modulus $2\pi\hbar$ in Q335	p $\{2\pi\}$ = 2 × p π	Double the π numerator
Quotient	$\lfloor p\pi / (2 \times p\pi) \rfloor = 0$	Integer: 0 (ground state quantum number)
Remainder	$p\pi - 0 \times 2p\pi = p\pi$	Exactly π in Q335
Physical remainder	$p\pi / p\{2\pi\} = 1/2$	The Maslov correction — exactly 1/2
Zero-point energy	$E_0 = \hbar\omega \times (\text{remainder}/2\pi) = \hbar\omega/2$	Exact

Example 3: Berry phase for Z_2 topological insulator

Step	Operation	Value
Geometric phase	$\gamma = \pi$ (protected by time-reversal symmetry)	Half the modulus
In Q335	$p\gamma = p\pi$	π numerator
Modulus 2π in Q335	p $\{2\pi\}$ = 2 × p π	Double π numerator
Quotient	$\lfloor p\pi / (2p\pi) \rfloor = 0$	Integer: 0 (winding number)
Remainder	$p\pi - 0 = p\pi$	Exactly π
Physical remainder / modulus	1/2	Topological — either 0 or 1/2, nothing in between
Observable	$\sigma_{xy} = (e^2/h) \times (\gamma/\pi \bmod 2) = e^2/h$	Quantized Hall conductance

In all three examples: The quotient is an exact integer. The remainder is an exact ratio of Q335 numerators. No rounding occurs. The physical observable (zero-point energy, Hall conductance, CP violation strength) is determined by the remainder alone. The Q335 framework makes the integer structure computationally exact.

APPENDIX R: THE COMPLETE SEARCH SPACE — WHAT HAS AND HAS NOT BEEN TESTED

Search Type	What Was Tested	Scope	Result	What Remains Untested
PSLQ linear (MATH-6)	82 constants against 20-constant basis	maxcoeff ≤ 10,000	82/82 null	Nonlinear relations; coefficients > 10,000; constants not in basis
PSLQ linear (PHYS-10)	17 SM ratios against 35-constant basis	maxcoeff ≤ 10,000	17/17 null (× 3 pool sizes = 51 null)	Same
PSLQ residual	$\alpha^{-1} - 114\zeta(3)$ against 11 constants	maxcoeff ≤ 10,000	6/6 null	Larger pools; higher maxcoeff
Modular scan (single modulus)	17 targets × 34 moduli	d ≤ 30 for remainder fractions	No clean hits	Products of moduli; scale-dependent moduli
Modular scan (gauge-specific)	3 couplings × 3 group-specific moduli	At M Z only	No clean hits	Across all energy scales
Products of basis constants	Not tested	—	—	~500 pairwise products as moduli
Self-referential moduli	Not tested	—	—	$f(\alpha)$ as modulus for α
Scale-dependent scan	Not tested	—	—	Modular scan at 100+ energy scales
Nonlinear relations	Not tested	—	—	$\alpha = f(\pi, \zeta(3), \ldots)$ for nonlinear f
Multi-constant moduli	Not tested	—	—	$(a\pi + b\zeta(3))$ as composite modulus

Coverage estimate: The linear and single-modulus searches cover approximately 5-10% of the plausible search space. The remaining 90% consists of products, nonlinear relations, scale-dependent structures, and self-referential moduli. The null result is a bound on the simplest possibilities, not a proof of non-existence.

APPENDIX S: THE PATTERN — INTEGERS, TRANSFORMATION LAWS, AND MEASUREMENTS

The complete structure of the HOWL series in one table.

Category	What It Contains	Examples	How Many	Status
Pure integers	Topological invariants, quantum numbers, counting	$n \in \mathbb{Z}$ (winding), N c = 3, N gen = 3, $c_2 \in \mathbb{Z}$	Countably infinite	Complete — the integers are known
Exact rationals	SM coefficients from diagram combinatorics	$A_1 = 1/2$, $b_0 = 41/10$, 197/144, 218/115	~50 named rationals in SM	Complete through 3-loop
MATH-2 transcendentals	Constants from convergent rational series	π, ln(2), $\zeta(3)$, $\zeta(5)$, $Li_4(1/2)$	34 in Q335 basis	Complete for known QED
MATH-3 transcendentals	Constants requiring extended methods	$K(k^2)$, $E(k^2)$, higher ζ values	~12 additional	Partial — 4-loop wall
Transformation laws	Functions connecting measurements at different depths	QED series, VP running, Koide, β functions	~10 identified	Known for EM sector; unknown for most SM parameters
Moduli	Symmetry/topology-determined periodicities	2π, $2\pi\hbar$, 1 (CS), $2\pi/N$	5 formal domains	Known in 5 domains; unknown for SM couplings
Selection principles	What determines the remainder	Energy minimization, critical amplitude, perturbative series	3 identified (θ, Koide, α)	Known for 3 parameters; unknown for remaining 17
Measured rationals	Values supplied by the universe	α^{-1}, m e, m μ, $\sin^2\theta$ W, α s, …	18 remaining (after θ QCD = 0)	Not derivable without new laws
The gap	Parameters with no known law or modulus	Yukawa couplings, CKM angles, m H	17 (or 16 with Koide)	The open problem

The program: Move parameters from "measured" to "derived" by identifying the transformation law, the modulus, and the selection principle for each. Where these are known (θ, Koide, α), the derivation follows. Where they are unknown, the parameter remains measured. The PSLQ null (139 tests) bounds where the modulus is NOT. The open directions (Appendix P) define where to search next.

[**HOWL-PHYS-10-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/
PHYS-10-2026

[**HOWL-MATH-4-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH
MATH-4-2026

[**HOWL-PHYS-7-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-7-2026

[**HOWL-PHYS-9-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-9-2026

[**HOWL-PHYS-5-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-5-2026

[**HOWL-MATH-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/MATH
MATH-2-2026

[**HOWL-PHYS-6-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-6-2026

[**HOWL-PHYS-8-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/PHYS/H
PHYS-8-2026